

1. **Define Function $F(x)$**
 - o Input: A number x
 - o Output: Return the value of the expression $x^2 - 4x - 10$
2. **Define Derivative Function $dF(x)$**
 - o Input: A number x
 - o Output: Return the value of the derivative of $F(x)$, which is $2x - 4$
3. **Define Procedure for Newton-Raphson Iteration**
 - o Input: x_0 (initial guess for the root), $noit$ (maximum number of iterations), acc (acceptable accuracy)
 - o **Check if the initial guess is already a root**
 - If $|F(x_0)|$ is less than or equal to acc , print that x_0 is the root and stop the program.
 - o **Initialize variables**
 - Set x_m (updated guess using Newton-Raphson formula)
 - Set iteration counter n to 0.
 - o **Loop until root is found or max iterations reached**
 - Repeat:
 - Compute new guess $x_m = x_0 - F(x_0) / dF(x_0)$ using Newton-Raphson formula.
 - If $|F(x_m)|$ is less than or equal to the acceptable accuracy (acc), stop the loop.
 - Update x_0 to x_m (use the new guess for the next iteration).
 - Increment the iteration counter n .
 - Continue the loop until either $|F(x_m)|$ is less than acc or the number of iterations reaches the limit $noit$.
 - o **Print the result**
 - Output the root x_m and the number of iterations n .
4. **Main Program**
 - o Set the initial guess $x_0 = 5.0$
 - o Set accuracy $acc = 0.00001$
 - o Set maximum iterations $noit = 30$
 - o Call the Newton-Raphson procedure with these inputs.